| MUHAMMAD IMRAN | BT201 Ecology, Biodiversity & Evolution-II |
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| MCQs | ALL IN ONE FILE |

| GRAND QUIZ | QUIZ 1 | MID TERM | QUIZ 2 | FINAL TERM |
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# GRAND QUIZ MID TERM

1. The complete and published genomes of plant by June 2005 _____ **4**
2. The complete and published genomes of Apicomplex by June 2005 _____ **6**
3. The complete and published genomes of Entamoeba by June 2005 _____ **1**
4. The complete and published genomes of archaea by June 2005._____**47**
5. The complete and published genomes of animal by June 2005_____ **16**
6. The complete and published genomes of bacteria by June 2005 _____ **211**
7. The complete and published genomes of aves by June 2005_____ **1**
8. The complete and published genomes of bacillariophyta by June 2005 _____ **1**
9. The complete and published genomes of pisces by June 2005_____03
10. The complete and published genomes of mammalia by June 2005._____ **5**
11. The complete and published genomes of nemaoda by June 2005._____**2**
12. The complete and published genomes of tunicata by June 2005._____**1**
13. The complete and published genomes of insecta by June 2005._____**4**
14. The complete and published genomes of eukarya in total by June 2005 _____ **47**
15. Saccharomics is the study of all the …… in the cell _____ **carbohydrates**
16. …..elements that can 'jump' into the DNA of drosophila for introduction of foreign DNA. _____ **p**
17. Species richness is generally highest in …. latitudes, and then decreases to the north and south, with zero species in the Arctic regions. _____ **tropical**
18. Brittle star belongs to which phylum _____ **Echinoderms**
19. Leeches belong to phylum _____ **Annelida**
20. A ….. grouping consists of various species that lack a common ancestor _____ **polyphyletic**

21. A …. is a group of species that includes an ancestral species and all its descendants _____ **clade**
22. One of the following option is not an example of Ex situ conservation of species _____ **natural parks and nature reserves**
23. the smallest units that can sustain life in isolation from all but atmospheric surroundings is known as _____ **Ecosystem**
24. Habitat destruction is a major cause for _____ **biodiversity loss**
25. Class Clitellata belongs to phylum. _____ **annelida**
26. The template nucleic acid used for translation is _____ **mRNA**
27. …… uses DNA and other molecular data to determine evolutionary relationships _____ **Molecular systematics**
28. Adult Echinoderms are _____ **radially symmetry**
29. Australopithecus afarensis was first bipedal hominid found in east Africa about _____ **3.0-3.2 MYBP**
30. Phanerozoic consists of _____ **all of them**
31. In an individual organism, is the state of possessing different alleles at a given locus in regard to a given character? _____ **Heterozygosity**
32. The puffer fish, takifugu rubripes one-eighth of …… genome _____ **homo sapiens**
33. The kudzu plant was introduced to the….. from Asia in the late 1800s as an ornamental plant _____ **America**
34. Gymnosperms have _____ **naked seeds**
35. large areas of protected land for conservation of biodiversity and the traditional life of the tribals living there. _____ **Biosphere reserves**
36. Earliest fossils dated to 200 million years ago are _____ **Turtles and tortoises**
37. RNA shows enzymic ( catalytic ) properties called ------_____. **Ribozymes**
38. Genetic diversity**,** means that the population contains most of the possible alleles for each particular _____ **gene locus**
39. Zoologist who study birds are called _____ **Ornithologist**
40. Some expert's estimate that around 30% of all species on earth will be extinct by _____ **2050**
41. Different individuals carry different genetic information, The sum of the variation of an interbreeding population is called _____ **Gene pool**
42. sequencing using mass spectrometry, in which masses are estimated for ….. generated from proteolytic digests of the protein _____ **peptides**
43. how the birds are designed to fly? _____ **Three toes pointing forward and one toe pointing backward**.
44. Cynobacteria introduced significant amount of _____ **o2**
45. Ticks belong to phylum _____ **Arthropoda**
46. arctic communities include just a …..species packed into each cubic meter _____ **few**
47. ballooning occur in _____ **Chelicerata**
48. which of the following system is found in Echinoderms? _____ **system of internal skeleton**

**49.** ….. species are those species of plants and animals found only in a particular area and not found anywhere else. _____ **Endemic**

**50.** Which of the following structure is present in all the chordates_____ **Notochord**

**51.** The protease used in Mass spectrometry is _____ **Trypsin**

**52.** the study of ecology in urban areas and the interactions between organisms is called …. ecology _ _____ **urban**

**53.** One of the following is not the consequences of deforestation _____ _____ **Decrease in the level of carbon dioxide in the atmosphere**

**54.** Notochord is _____ **flexible**

**55.** Spider belong to subphylum _____ **Chelicerata**

**56.** human genome project the assembled the ….billion bp of DNA, _____ **3.12**

**57.** the …..is a function of both the number of different species in the community (species richness) and their relative abundances _____ **Species diversity**

**58.** dinoflagellate belong to _____ **Protozoans**

**59.** larvae of Echinoderms are _____ **bilaterally symmetrical**

**60.** contribution of an individual's genotype to the next generation_____ **Fitness**

**61.** are the plants found in a particular area _____ **Flora**

**62.** Large scale areas of similar vegetation and climatic characteristics are known as _____ **Biome**

**63.** The study of transcriptomes is part of …..genomics _____ **functional**

**64.** Trochophore larvae develop, which are remarkably similar to the_____ **Molluscs**

**65.** Features that have no apparent function _____ **Vestigial Structures**

**66.** …..is the ordered division and naming of organisms _____ **Taxonomy**

**67.** Some species known as ….. have unusually strong impacts on community structure, preserving the balance of the community or even making its existence possible_____ **keystone**

**68.** Humans have been evolved about …. Year ago._____ **5 million**

**69.** Seeds can be maintained for decades or even centuries if the conditions are controlled _____ **<5% humidity**

**70.** Crabs have ……. Body symmetry _____ **Bilateral**

**71.** Breeding in annelids occurs in _____ **spring**

**72.** …….. may also be used to determine the amino acid sequence of a protein.___ _____ **Mass spectrometry**

**73.** In bioinformatics HMM stand for _____ **Hidden Markov Models**

**74.** The molecular clock says ….. million years ago the human-chimp line split. _____ **~ 5**

**75.** Who is not an invertebrate? _____ **amphibians**

**76.** property of a human society in which ecosystems (including humans) are managed such that the conditions supporting present day life on Earth can continue. Is known as _____ **Sustainability**

**77.** The more variation, the better the chance that at least some of the individuals will have an …… that is suited for the new environment. _____ **allelic variant**

**78.** Pharyngeal gill slits_____ **are unique chordate characteristic**

**79.** Which of the following is a typical feature of prokaryotic cell _____ **Absence of nucleus**

**80.** the silver carp, eats…, a key food for many native fish species _____ **plankton**

**81.** Evolutionary fitness is a measure of -_____ **Reproductive fitness**

**82.** The genome size of living organisms ranges across …. orders of magnitude,_____ **09**

**83.** The laborious ..technique has now been replaced mostly by sequencing using mass spectrometry__ _____ **edman degradation**

**84.** According to the International Union for Conservation of Nature (IUCN), globally about …..of all known species are threatened with extinction_____ **one third**

**85.** Which of the following are characteristics of prokaryotic cell _____ **all of the above**

**86.** sea squirt is a model and belongs to a group the _____ **Urochordata**

**87.** Oxidizes amino acids, fatty acids and alcohol known as _____ **Peroxisome**

**88.** the velvet worm, , traditionally seen as a missing link between the ___ **arthropods and annelids**

**89.** Loss of genetic diversity can imply lack of …and premature extinction _____ **evolution**

**90.** ….. is the evolutionary history of a species or group of related species _____ **Phylogeny**

**91.** The spread of Homo out of Africa is presumed to have taken place about …by Homo erectus ._____ **1.5 MYBP**

**92.** coral reefs which are biodiversity hotspots will disappear in _____ **20-40 years.**

**93.** 18th century, ….published a system of taxonomy based on resemblances _____ **Carolus Linnaeus**

**94.** Ancient order inhabit few small islands of new Zealand _____ **Rhynchocephalia**

**95.** Invertebrates Makes up …. of all Animal species _____ **95%**

**96.** The size of puffer fish, ….of the human genome _____ **one-eighth**

**97.** One of the following is not example of positive interaction among the microbes ___ **Amensalism**

**98.** Gregor Mendel's experiments with plant hybridization led to his establishing _____ **laws of inheritance**

**99.** Cockroach has ……. Body symmetry_____ **bilateral**

**100.** arabidopsis employs the …. photosynthetic pathway. _____ **C3**

**101.** the stage of parasitic protozoa that actively feed and multiply are frequently called_____

**102.** A taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species _____ **Barcoding**

**103.** ecologists use to describe the composition of a community are species richness and …diversity _____ **species**

**104.** which is not a foundation species_____ **green algae**

**105.** . Black carp eat…., limiting their availability for native fish and damaging shellfish populations _____ **mussels**

**106.** Haldane (1929): gave the _____ **"Hot, dilute soup" theory**

**107.** Genetic diversity is the variation at the level of _____ _____ **both at individual genes, and In a population,**

**108.** …… carp illustrate the potential impact of invasive species _____ **Asian**

**109.** one of the following is the not part of maintaining balance of the ecosystem__ _____**protecting food crops**

**110.** levels of studying are _____ **Population Cell/Organism Organelle Molecule Atom**

**111.** .is a foundation species that forms the basis of the kelp forests off the coast of california._____ **brown algae**

**112.** **intermediate disturbance hypothesis** suggests that communities with a …may have greater species diversity than communities _____ **intermediate disturbance**

**113.** Humans are members of the order _____ **Primates**

**114.** Genetic modification of plants is achieved by adding a specific gene or genes to a plant, or by knocking down a gene with…., to produce a desirable phenotype. _____ **RNAi**

**115.** Class Polychaeta is the class of Phylum _____**Annelida**

**116.** A valid clade is ---, signifying that it consists of the ancestor species and all its descendants. _____ **monophyletic**

**117.** Extra chromosomal self replicating circular DNA molecules present in bacteria are called;_**plasmid**

**118.** In leeches, salivary glands excrete _____ which prevents the blood from coagulating.__**hirudin**

**119.** ..... mutations are often lost from a population over subsequent generations_____ **Bad or lethal**

**120.** In which of the following animal sexual reproduction takes place?_____**cockroach**

**121.** A taxonomic unit at any level of hierarchy is called a _____**taxon**

**122.** Methanogens fall under the domain of;_____ **Archaea**

**123.** Sharp beak instead of teeth is characteristic of;_____ **Order Chelonia**

**124.** _____ Species are species that have been introduced into areas outside their native range and can cause—or have caused—harm in their new area._____ **invasive**

**125.** Vestimentiferans belong to phylum _____ **Annelida**

**126.** ….assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely _____ **Maximum parsimony**

**127.** Krill is the major food for _____.**whales**

**128.** Ecologists use to describe the composition of a community are _____richness and species diversity. _____**species**

**129.** Trypanosoma is spread by _____._____**tsetse fly**

**130.** On 26 June 2000, when US president _____, J. Craig venter of Celera genomics, and Francis Collins of the national institutes of health jointly announced that a working draft of the human genome had been completed._____ **B. Clinton**

**131.** Sea anemone has_____ body symmetry._____ **radial**

**132.** Who is not a model organism?_____ **hony bee**

**133.** --------- was probably the first hereditary molecule having the ability to copy itself._____**RNA**

**134.** Populations of endangered species are small and tend to lose;_____ **genetic diversity**

**135.** Group of interacting and interbreeding organisms is known as; _____**population**

**136.** Systematists have proposed the----------, which recognizes only groups that include a common ancestor and all its descendents._____ **phylocode**

**137.** Species richness is the number of different species in a particular _____.**community**

**138.** Assigning functions to genes and recognizing _____ sequences in a model genome can greatly benefit from comparison with a set of carefully chosen reference organisms at defined phylogenetic distances._____ **promoter**

**139.** The following are the major activities of plant breeding One of following gives the correct sequence of events in Plant breeding ___ **Collection of variation. Selection Evaluation Release**

**140.** The.., , traditionally seen as a missing link between the arthropods and annelids _____**velvet worm**

**141.** Centipedes belong to subphylum_____. **myriapoda**

**142.** The international human genome project initiated by the US national institutes of health and the US department of energy, was launched in _____-__.**1990**

**143.** Which one is an extinct group of arthropoda?_____ **crustacea**

**144.** Acritarchs are very likely the cysts of;_____ **algae**

**145.** _____ theory states that groups of similar organisms are descended from a common ancestor _____ **Evolutionary**

**146.** Seeds which are not of agricultural importance yet hold traditional importance; these seeds are kept in seed banks and are still maintained by some gardeners and farmers are known as;_____**Heirloom**

**147.** By june 2005, no fewer than prokaryotic organisms and 496 eukaryotes were the subject of ongoing genome sequencing projects. _____**730**

**148.** …..and wildlife can be conserved by setting up areas like Biosphere reserves Wildlife sanctuaries, National parks, etc. _____ **Forest**

**149.** ….. is the study of all low molecular weight cellular constituents. _____ **Metabolomics**

**150.** If for genetic modification genes of the species or of a crossable plant are used under control of their native promoter, then they are called_____ **cisgenic plants**

**151.** Aves are ----_____- , **both A and B**

**152.** The locomotary organ of echimnoderm are called ------._____ **Tube feet**

**153.** Antibiotic resistance gene rapidly spread within and between bacterial populations by transposons called ._____ **R plasmid**

**154.** Two million people died in potato famine in ---_____. **Ireland**

**155.** The international human genome project initiated by US was completion due in. _____**2000**

**156.** The proterozoic eon alone accounts for ---- of all geologic time. _____**42.5**

**157.** Haldane 1929 gave the ----_____ **Hot dilute soup theory**

**158.** CRF stands for-----_____**Conditional random field**

**159.** Wormlike and algae fossils come from ----- million years old rock in china. _____**700-900**

**160.** Homo sapien are believerd to have originated around how many years ago. __ **100,000-200,00**

**161.** Class squamata has ------.Backbones of _____**100-400 vertebrates**

**162.** The -----synthesize glucose using sloar or chemical energy which is broken down through respiration.

_____**Autotrophs**

**163.** By 2 billion years ago free oxygen was appearing in the atmosphere due to the activity of -------.

_____**Photosynthetic bacteria**

# QUIZ NO 01

1. **In protozoa era only ---------- are recognized.** _____**Bacteria**
2. **------- mutations are often lost from a population over subsequent generation.**

   _____**Bad or lethal mutations**
3. **Evolutionary fitness is a measure of ----------.** _____**Reproductive fitness**
4. **Coral reefs which are biodiversity hotspots will disappear in -------- year** _____**20-40**
5. **In the 18th century Carolus Linnaus published a system of taxonomy based on resemblance . two key features of his system remain useful today two part names of species and hierarchical classification.**_____ **Two part names for species and hierarchical classification**
6. **One of the following is not a part of maintaining balance of the ecosystem.**

   _____**Protecting food crops**
7. **Gregor Mendals experiments with plant hybridization led to the establishing.**_____ **Laws of inheritance**
8. **Haldane 1929 gave the ----------.**_____ **Hot dilute soup theory**
9. **---------- uses DNA and other molecular data to determine evolutionary relationship.**

   _____**Molecular systematic**
10. **In genome evolution gene number and the complexity of an organism-------.**

    _____**Are not strongly linked**
11. **Medicines from nature account for usage by -------- of the world population.** _____**80%**
12. **One of the following is not the consequences of deforestration.**

    _____ **Decrease in the level of CO2 in the atmosphere**
13. **Large scale areas of similar vegetation and climate characteristics is --------.** ____**Biome**
14. **If for genetic modification genes of the species or of a crossable plants are used under control of their native promoter then they are called --------- plants.** _____**Cisgenic**

# MID TERM

1- Humans are included in the order ----------._____primates
2- The field of biology specializing in storing and analyzing data -------. ____Bioinformatics
3- The possibility of working with molecules on the scale of few micrometers is called ----- .
_____micro ecology
4- First question asked when a species is bought for genome sequencing ----------
._____ What is the size of its genome
5- The study of transcripts that are present at any time in the cell are called --------
_____transcriptomics
6- Importants measures used to describes community composition are ---------
._____ Species richness and species diversity
7- Vertebrates makes --------- % of the animals . _____5%
8- The number of allelle in a sample ----------. _____Allelic richness
9- Mass spectromatry used to determine ---------
sequence._____ Amino acid sequence of protein
10- Oparin 1924 work ----------. _____Primeval soup theory
11- Quinine is obtained from quina it is used to treat ----------._____ Malaria
12- The common extension of genomics are except ---------. _____Vaccinomics
13- To assemble a genome from a series of sequence fragment require tremendous.
_____Computational power
14- The first genome sequenced completely was that of ----------. _____Hemofelues influenza
15- The ------- technique has now been replaced mostly by sequencing using mass
spectrometry._____ Edman degradation
16- Annelids do not have feeling style -------. _____Filter feeders
17- Modern bread wheat is a mixture of recombined genes from --------._____ 3
18- Gymnosperms means ---------. _____Naked seeds
19- Medicines from nature account for usage by --------- of the world population. _____80%
20- Habitat destruction is the majorcause ---------. _____Biodiversity loss
21- Coral reefs which are biodiversity hotspot will disappear in ----------. _____20-40 years
22- Plants found in a particular area is ----------. _____Flora

23. Gregor mendal eperiments with plants hybridization led to this establishing of -_
_____Law of inheritance
24. Genetic modefication of plant is achieved by adding a specific gene or genes to a plant or by knocking down a genewith to produce a desirable phenotype. _____RNAI
25. They posses hotochord in any period of their life --------. _____Chordata
26. One of the following is not the consequences of deforestration ----------.
_____Decrease in the level of carbon dioxide in the atmosphere
27. If for genetic modefication genes of the species or of a crossable plant are usedunder control of their native promotor then they are called --------- plants. __CISGENIC
28. The following are the major activities of plant breeding one of thefollowing , gives the correct sequence of events in plant breeding ----------.
_____Collection of variation , selection, evluation, release
29. One of the following is the NOT part of maintaining balance of the ecosystem. ---------
_____.Protecting food crops

30. Which one is not related to human genome project --------- three synteny _____analysis, phylogentic, foot printing, phylogenetic shadowing. Other than following

31. Species differ from foundation species in that they are more like to belong to ---- _____All of above

32. Which is not related to endosymbiosis ----------. _____Prokaryotic cell study

33. Which is not the part of horizental gene transfer----------. _____Conjugation, transformation, transduction. Other than these

34. Which one in not related to protozoa reproduction. _____ Parthenogensis

35. Taxonomy was established by ---------. _____CAROLUS LINNAEUS

36. Linus Pauling called science a way to find truth or logic , select correct. _____Truth

37. ---------- assumes that the tree that requires the fewest evolutionary events is the most likely._____ Maximum parismony

38. Annelids are not predators ----------_____

39. Not related to hardly weinburg theorem ----------_____

40. The field of biology specializing in developing hardware and software to store anfd analyze the huge amount of data being generated by life scientists -------. __Bioinformatics

41. There are some so called --------- to identify ecologically genes in incompletely genome. _____Pregenomic molecular approaches

42. The international human genome project initiated by except ----------. _____US national institutes of genetics

23- Comparative genomics is developing an increasing array of bioinformatics technique except ----------. _____Synteny shadow

24- Keystone species differ from foundation specie higher trophic level and top predator , tend to modefy their environment. _____All give above

25- Eukaraya have been evolved from since --------- million year ago. _____Two

26- Group consisting various species that lack a common ancestor called ----__polyphlyletic

27- Types of selection which favor more than two extreme phenotype is - _____ Disruptive selection

28- Which of the following is not mode of feeding of annelids ----------._____ Predators

29- A philospher concept of value of something independent of its value to anything else called -------._____ Intrinsic value

30- Krill is a major food for ---------. _____Whales

31- Most important measures ecologists use to describe composition of community are ---------. _____Species richness and species diversity.

32- Major technological developments of 1990 computing, communication, micro technology. _____All of above

33- ---------- are found in more than one copy in the genome. _____Paralogus genes

34- In Edman degradation technique study of all the proteins in the cell which is not related. _____Fragment pattern is not compared with genomic sequence

35- Which is not true about protein degradation. ------. _____Sequencing using mass spectrometry replaced by degradation

57. ……..is used to cleave the protein. _____Trypsin

58. Arabia thaliana's is……..._____ Fly

59. All peptide technique are…….. _____All

60. ……. Is used to predict the protein structures. _____Mass spectrometry

61. Trochophore resembles like …….. _____Larvae

62. Plasrnid size …….. _____2 to 200 bp

63. Humans differ from gorilla…….. _____bipedal

64. That's are transfer …….. _____Traits

65. Kimber Ella is …….. _____Animal.

66. Coral reef are disappeared due to …… _____Global Warming

67. Vertebrates are …….. _____5%

68. ……. The possibility of working with molecules on the scale of a few micrometers, _____Micro ecology

69. Which institute was not involved in human genome project? …_____ U.S...Center of health

70. Krill component of plankton major food for ----------. _____Whales

71. Muscular pharynx of annelids is called ---------. _____Eversible proboscis

72. Segmentation is confined to rear portion of annelids called --------. _____Opisthosome

73. Zoologist that study birds known as ---------. _____Ornithologist

74. Birds are ----------. _____Homeothermic

75. Turtles and tortoise belong to order ----------. _____Chelonian

76. Biodiversity is the result of 3.5 billion years of ----------._____ Evolution

77. 80% of human food supply comes from -------- kinds of plants. _____ 20

78. ---------- is the application of biotechnology and molecular biology. __Molecular breeding

79. Edward Wilson described the ----------._____ Acronym

80. ---------- of all species on earth will be extinct by 2050. _____ 30%

81. -------- ecosystems are the most threatened ecosystems. _____ Freshwater

82. Over ------- of global fisheries are being overfished. _____25%

83. Ayubia national park is the first --------- in Pakistan. _____Reserve forest

84. Effective population size is much smaller than ------ population size. _____Census

85. Seeds can be maintained in seed bank at ----------. _____ 20℃

86. CITES the convention in international trade in endangered species set up in --------. 1988

87. Plasmids size ----------. _____2-200bp

88. Conjugation, transformation and transduction are --------- gene transfer. _____Horizontal

89. Oparian 1924 gave the --------. _____Primeval soup theory

90. RNA shows enzymic properties called ------. _____Ribozymes

91. Analogies are also known as ----------. _____Homoplasis

92. ---------- years ago free oxygen was in atmosphere. _____2 billion years

93. Human order primates consist of -------- species. _____180

94. ---------- groups of organisms by common descent. _____Cladistics

95. All known protozoic animals wre ----------._____ Softed bodied

96. Malaria is caused by plasmodium technically a member of phylum ---------._____ Sporozoa

97. Lyme disease caused by ----------. _____Tick

98. Larva of annelids trochophore similar to the ----------._____ Mollusks

99. Gut tissues form an organ ----------. _____Trophosome

100. Trophosome play a role in ----------. _____Nutrition

101. ---------- is a small ball above the tube foot. _____Ampullae

102. Vertebtrates are evolved from ---------- 540 million years . _____chordates

103. Chordates are bilaterian animals that belong to clad of animals known as ----------. _____Deterostomia

104. The study of traditional reptile orders ----------._____ Herpetology

105. Humans use ---------- species for food clothing and shelter. _____40,000

106. ---------- of all mammals will be extint in 20 years. _____20%

107. The sum of the variations of an interbreeding population is celled ---------- gene pool

108. Homology is similarity due to ---------- b/w wolf and coyote. _____Shared ancestory

109. Analogy is similarity due to --------- in similar condition. _____Convergent evolution

110. A ---------- is a group of species that includes on ancestral species._____ Clade

111. ---------- an anmal from latest proterozoic rocks in Russia. _____Kimberella

112. Three types of homology, 1-structural 2-developmental 3-genetic _____ All

113. Homologus structures ----------. _____Same structure & different functions

| MUHAMMAD IMRAN | BT201 Ecology, Biodiversity & Evolution-II |
|---|---|
| LESSON NO 51 TO 61 | QUIZ NO 02 |
| 77 MCQs | |

1. Light within this band, which is called photosynthetically active radiation, or PAR, accounts for about ….% of the total energy content of the solar spectrum at sea level. _____45
2. Light within photosynthetically active radiation band ranges from ….to…. nm._____ 400-700
3. In 1997 ……..% of the global human population was spread across Africa_____12.8
4. …….. Realized life could not all be ordered in a single chain._____ Lamarck
5. Co-operative groups of females in African lions are called ……_____ prides
6. ……. Showed that the claims of undeniable catastrophes in the record where much exaggerated ._____ Lyell
7. …… found a significant positive correlation between population size and genetic diversity in populations of chihuahua spruce._____ Ledig
8. The type of survivorship in which there is fairly constant probability throughout the life is called ……._____ type II
9. In 1997 ……. % of the global human population was concentrated in Asia._____61.4
10. …… explored the unity of parts and their patterns of development-skeletons in different vertebrates._____ Hilaire
11. The relationship between photon flux density and plant photosynthetic rate is called _____photosynthetic response.
12. ….. is not involved in exploring the extent and source of morphological vibration in plant population including both the influences of environment and genetics._____ Kerner
13. One of the most fundamental social interactions between individual take place during._____ sexual reproduction.
14. Photosynthetic plants and algae use light, of wavelengths between….. and …..nm, as a source of energy _____ 400 and 700
15. The turnover of water in the various reservoirs of the hydrologic cycle ranges from only …….days for the atmosphere to 3,100 years for the oceans._____ 9
16. The first biologists to conduct thorough studies of phenotypic and genotypic variation and to incorporate experiments in their studies, focused on_____ plants
17. Only ……. Is a desert plant genus _____Encelia
18. ……..identified four basic body plans for animals– vertebrates, molluscs, arthropods, and radiata (jellyfish & others)_____ Cuvier's
19. The nature of …….assemblages is being used to assess the "biological integrity" of freshwater communities._____ fish
20. …… reported that soapberry bug populations living on native and introduced host plants have undergone natural selection for traits that favor their survival and reproduction on particular host plant species_____ Carroll
21. ……..are the most tropically diverse organisms in the biosphere._____ Bacteria

22. He cited teeth in fetal baleen whales_____ Lamarck

23. Kelp forests are generally limited to areas where temperature ranges from about …….. $^0$C to ……$^0$C _____ 10°to 20$^0$C

24. The salinity of lakes, which ranges from very dilute waters to over …..‰, is much more variable than that of the oceans_____200

25. The …..distribution of a population reflects its history of survival, reproduction, and potential for future growth._____ age

26. Water potential, osmotic pressure, and vapor pressure deficit can all be measured in _____pascals

27. Detritivores feed on dead ……material, which is even lower in nitrogen than living plant tissues._____ plant

28. …… showed that island populations generally include less genetic variation as compare to mainland population._____ Frankham

29. camel and……, use similar mechanisms to survive in arid climates._____ saguaro cactus

30. Plants and ectothermic animals regulate their heat exchange through their _____ Both

31. niches of Galapagos finches are largely determined by their_____ feeding

32. In 1997, …….. % of the global population was spread across oceanic island _____0.1

33. Of the water in the biosphere, the polar ice caps and glaciers contain_____2%

34. Optimal foraging theory attempts to model how organisms ……as an optimizing process_____ feed

35. expanded the logistic model of population growth to represent interspecific competition.__ _____Lotka

36. The turnover of water in the various reservoirs of the hydrologic cycle ranges from only 9 days for the atmosphere to ……years for the oceans _____ 3,100

37. Schoenerused replicated field experiments to study natural selection for changes in morphology in Anolis ……populations_____ lizard

38. The trophic diversity of……., which is critical to the health of the biosphere, can also be used as a tool to address some of our most challenging waste disposal problems _____ bacteria

39. low mortality among the young but high mortality among older individuals is …… survivorship_____ type I

40. Since the environment is steadily changing for….., some plants and animals will become extinct because they're no longer adapted to the environment they're in_____ Lyell

41. Reported that the difference in growth form were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants._____ Kerner

42.a state of reduced metabolism that may last several months_____ tropor

43.Kelp forests are found mainly at …..latitudes_____ temperate

44.Kelp beds share several structural features with ……. Forests. _____ terrestrial

45.The relationship between food density and animal feeding rate is called the_____ functional response.

46.Leafhoppers are an example of …….. competition._____ Intraspecific

47.He added the concepts of the fundamental niche, the physical conditions under which a species might live in the absence of other species,_____ Hutchinson

48.In 1997, …….. % of the global population was spread across Australia_____0.3%

49.a state of low metabolic rate and lowered body temperature, when food is scarce and night temperatures cold_____tropor

50.C4 and CAM plants are more …..in their use of water than are C3plants. _____ efficient

51.……. found that higher heterozygosity (genetic diversity) was associated with lower rates of population extinction_____ Saccheri

52.instead of dying out (in most cases, anyway) ancient animals and plants evolved into different forms that continue today was by _____ Lamarck

53.The first biologists to conduct thorough studies of phenotypic and genotypic variation of plants was _____. Bonnier

54. ____ studies provided evidence for genetic differences among populations._____Turesson's

55.Population growth stops when populations reach a maximum size called the_____. carrying capacity

56.On land, the tendency of water to move from organisms to the atmosphere can be approximated by the vapor pressure deficit of the_____ air

57.The water potential of pure water, which by convention is set at_____ 0

58.Population densities in different regions vary from less than 1 person per square kilometer to nearly …..persons per square kilometer_____ 1,000

59.…….may enter a state of torpor_____ Hummingbirds

60.One of the best predictors of rm is_____ body size

61.The ……reflects the environmental requirements of species _____ niche

62.Plants appear to allocate ……to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. _____ energy

63.Chemosynthetic autotrophs, which consist of a highly diverse group of chemosynthetic_____ bacteria

64.Of the water in the biosphere fresh water is less than ……% _____ 1

65. ……replicated field experiments to study natural selection for changes in morphology in Anolis lizard populations. _____Warhelt

66. Reef-building corals are limited to areas with temperature of about __ C to ___C. _____18-29

67. _____factors affect the population growth, _____both

68. _____ replicated field experiments to study natural selection for changes in morphology in Anolis populations. _____Losos

69. The niche of salt marsh grass is determined by _____. _____Tidal levels

70. In 1997, _____ % of the global human population, was spread across South America.___ (5.7)

71. Photosynthetic plants and algae ……as a source of carbon and light from the sun _____ CO2

72. Both seaweeds and reef-building corals grow only in _____ waters. _____ surface waters

73. In 1997, about _____ billion people of the global human population, were concentrated in Asia._____3.5

74. Self-thinning occurs in _____ as a competition for their survival._____ Plants

75. The type of survivorship having high mortality among the young and low mortality among older individuals is_____. _____type III

76. Of the water in the biosphere, the oceans contain _____% water_____97

77. ____ replicated field experiments to study natural selection for changes in morphology in Anolis populations._____ Schoenerused

78. ……are the most trophically diverse organisms in the biosphere._____ Bacteria

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1- -------- factors effect the population growth. Abiotic, biotic _____ both

2- The physical environment limits ------ distribution of species. _____Geographic

3- Community changes during succession include ------- in species diversity. ___Increase

4- A food web summarizes the ---------- relations in a community._____ Feeding

5- Percentage of oceanic water on earth is ----------. _____97%

6- Leaf hoppers are an example of --------- competition. _____Intra specific

7- Secondary forest succession on the piedmont platean takes about --------. Years. ___150

8- A species that is dependent on other organism is called as --------. _____Obligate

9- The total amount of energy fixed by all the autotrophs in the ecosystem is ---------- production. _____Gross

10- On islands and habitat patches on continents, species richness --------- with area. _____Increase

11- Plants and ectothermic animals regulates their heat exchange through their ----------. Morphology & behavior _____ both

12-**---------- studies provided evidence for genetic differences among populations.** _____Tureson,s

13-**C4 and CAM plants are more ---------- in their use of water than C3 plants.** _____Efficient

14-**---------- use organic molecules.** _____Heterotrophs

# FINAL TERM

1. **On Island and habitat patches on continents ----------.** _____Species richness increase with an area and decrease with isolation.
2. **Succession in the intertidal zone require --------.** _____1-3 years
3. **Highest rate terrestial primary production occur under ------.** _____Warm and moist conditions
4. **-------- in physical and chemical conditions across aquatic and terrestial environment can account for significant portion of the diversity among planktonic algae and terrestial plants --------.** _____Heterogeneity
5. **The refuges that promote the persistance of hosts and prey include ---------.** _____Secure places to which the exploiterhas limited access
6. **The environment limits population growth by -------.** _____Changing birth and death rate
7. **An environment in which resources are patchy also fosters--------.** Clumped distribution
8. **Eusoctalty is -------- .** _____more complex level of social behavior which is considered to be the pinnacle of social evolution
9. **The most familiar relative measures of watercontent of air is --------.___** relative humidity
10. **The rate at which organism can take in energy is ------- either by external or internal constraints --------.**_____ Limited
11. **The earth ecosystem interacting with physical environment as a whole maintained a steady state system intermediate in the flow of energy between the high input of the sun and thermal sink os specie is called ----------.**_____ Biosphere
12. **Ist multi cellular organism found in China as carbonaceous filament were --------** .____ 1.8 billion years
13. **Multicellular algae appear as carbonaceous filament from -------- old rock of china.** _____1.8 billion years
14. **Human is in which order ----------.**_____ Primates
15. **Maximum parsimony assume that the -------- is the most likely the.** _____Tree that require the fewest evolutionary events
16. **Oparian in 1924 gives ----------.** _____Primeval soup theory
17. **In 18$^{th}$ century Carlus Linnaeus published a system of taxanomy on resemblance------** _____. Two parts names of species and hierarchiel classification
18. **One of the following is not example of horizental gene transfer ---------.**____ Conjunction
19. **Which one of the following is not one of the positive interactions on microbe-------.** _____Amensalism
20. **Not positive interaction ----------.**_____ Competition
21. **---------- type of natural selection which favor more than two extreme phenotype is --------** .____ Disruptive selection

22. Which one of the following is not one of the advantages of in situ conservation ----------. _____The species have less space
23. The advantage of in situ conservation except ----------. _____The specie will not evolve in their environment
24. Which of the following is not common extension of omics --------._____ Vaccinomics
25. Verebrates make ---------- % of total animals. _____5%
26. Gymnosperms means ----------._____ Naked seeds
27. Protozoa reproduce except ----------. _____Parthenogenesis
28. Number of alleles in a sample population is called ----------. _____Allelic richness
29. One of the first question that is asked when a species is considered for whole genome sequencing is ----------. _____What is the size of its genome
30. Modern bread wheat is a mixture of recombined genes from ---------- different wild species . _____Three
31. Quinine is a drug used to treat ---------. _____Malaria
32. Annelids are not ----------. _____Predator
33. Species differ from foundation species is that are likely to belong ----------. _____Higher trophic levels
34. The ------- technique has now been replaced by sequencing using a mass spectrometry ---------._____ Laborios edman degradation
35. The first genome sequenced completely was that of -------._____ Homophilus influenza
36. To assemble a genome from a series of sequences requires tremendous ----------. _____Computational power
37. Community structure is measured by ----------. _____Species richness and diversity
38. A resilient community or ecosystem may be ---------- disrupted by distrubance but quickly return to former state._____ Completely
39. The ability to bounce back after disturbance is called ----------._____ Resilience
40. --------- demonstrated that the influence of consumers on community structure depends upon their feeding preferences, their local population density and the relative competitive abilities of prey species . _____Jane lubchenko
41. Notochord is present throughout life oe at least during embryonic stages in ----------. _____Chordates
42. Possibility of working with molecules on the scale of few micrometers is --------- ._____ Micro technology
43. Phylogenetic trees are evaluated in order to find the relationship among ----------. _____Species , phylum, family. All of given
44. Field experiment has ---------- points._____ Four
45. There are so called ---------- that may be used to identify ecological important genes in incompletely characterized genome. _____Pre-genomic molecular approaches
46. ---------- are using these new toools to study large scale dynamic ecological phenomenon such as international variation. _____Ecologists
47. The local population is density of the Glanville Fritillary --------- Meliteacinxia , is lower on larger and on isolated habitat patches._____ Butterfly
48. ---------- cells are combination of proteins synthesing system. _____Eukaryotes
49. Sonoran desert plant distribution map clearly on the soils of different ages and form a mosaic that closely matches soil mosaic. _____Vegetative

50. --------- occur on nearly exposed geological substrate not significantly modified organism. _____Primary succession
51. Supporting evidence for endosymbiosis except ------- comes from studies of living. Prokaryotic _____cells
52. ---------- can be used to detect long term ecological changes. _____Repeat photography
53. Who worked on the developing of paleontology --------. _____Smith,buckland, curvier, brongniart
54. A food ---------- summarizes the feeding relation in a community. _____Web
55. -------- proposed the feeding activities of few species have inordinate influence on community structure. _____Robert Paine
56. Hardy Weinburg principle conditions necessory to maintain constant alleles frequences in a population except ---------. _____Immigration
57. ---------- indepently expanded the logistic model of population growth to represent interspecific competition._____ Lotka Volterra
58. --------- explored the extent and sources of morphological variation in plant population including both the influence of environmrent and genetics. __Clausen, Keck, and Hiesey
59. The validity of field experiments depend upon several design features except ---------- ._____ Knowledge of final conditions
60. Populations include --------- variations among individuals which are significant aspect of population structure. _____ Genetic and phenotypic effects
61. ---------- soon found that even these so called simple communities included vary complex feeding relations. _____Charles Elton 1927
62. If for genetic modefication genes of the species or of a crossable plant are used under control of their native promoter, then they are called --------- plants. _____Cisgenic
63. A philosphical concept of value of something independent of its value to anything else is called ----------. _____Intrinsic value
64. Mass spectrometry is replaced by ----------._____ Edman Degradation
65. Which of the following is an example of negative interaction. _____Amensalism
66. The Noble Prize winning scientist Linus Pauling aptly described as the ---------. _____Search for truth
67. Study of transcripts that are present in an organism any time is called ------. _____Transcriptomics
68. Mass spectrometry is used to determine the sequence of --------. _____Amino acids
69. Which of the following is not part of human genome project----------. _____US department of medicines
70. Comparatvie genomics is developing an increasing array of bioinformatics techniques, which of the following is not done. _____ Synteny foot printing
71. The bark of this tropical tree contains a drug called quinine is used in the prevention of ------. _____Malaria
72. Medicines from nature account for usage by ------- of the world population. _____ 80%
73. Habitat destruction is major cause for ----------. _____Loss of biodiversity
74. Coral reefs which are biodiversity hotspots will disappear in -------- years. _____20-40
75. Plants founded in particular area is ----------. _____Flora
76. Georgemendal,s Experiments with plant hybridization led to his establishing ----------. _____Laws of inheritance

77. Genetic modefication of plants is achieved by adding a specific gene or gene to a plant or by knocking down a gene with ----------. _____RNAI
78. The following are major actvities of plant breeding , one of the following is correct sequence of plant breading ----------. _____Collection of variation-selection- evaluation- release (CSER)
79. One of the following is not the correct sequence of deforestration ---------- _____. Decrease in the level of carbon dioxide in the atmosphere
80. One of the following is NOT the part of nmaintaing balance of the ecosystem .-------- _____-. Protection of food crops
81. In genome evolution gene number and complexity of an organism ---------- _____-. Are not strongly linked
82. Gene diversity is the variation at the level of ---------- . _____both at individual gene and in a population
83. Haldane 1929 gave the ----------. _____Hot , dilute soup theory
84. Not example of Ex-situ Conservation ----------. _____Natural park and natural reserves
85. One of the following a population over subsequent generations----------. _____Bad or lethal mutation
86. The fossil record is relatively incomplete for several reasons , which one is not true statement.----------. _____Soft tissues are mostly preserved
87. Annelids dot not have feeding style --------. _____Filter feeders
88. Annelids are not -------. _____Predators
89. Evolutionary fitness is the measure of ----------. _____Reproductive fitness
90. ---------- uses DNA and other molecular data to determine evolutionary relationships. _____Molecular systematics
91. The number of allele in a sample ----------. _____Allelic richness
92. Plant complete when --------- nuterients are less. _____More intensively
93. The process of synthesis of protein is called ----------. _____Translation
94. Conjugation is a process of ----------. _____Sexual reproduction
95. Animals eat other animals are called -------. _____Predators
96. ----------group have common ancestor. _____Clade
97--------- could not all be ordered in a single chain._____ Life
98---------- arises from interaction b/w organism and environment._____ Evolution
99--------- identified the ------- basic body plans for animals. _____4
100- Smith, Buckland and Curvier himself were developing --------. _____Paleontology
101- development proceed up the tree of life. All animals begin as eggs._____ Von Baer
102- oceans contains ---------- of water. _____97%
103- the turn over of water of the hydrological cycle ranges from on ------ days for the atmosphere. _____9
104- ------- form the largest continuos environment on the earth. _____ Oceans
105- Photosynthetic organisms restrict to ---------- zone. _____Euphotic or epipelagic
106- Oceanic temperature much more stable than -------- temperature. _____Terrestrial
107- Highest productivity occurs along --------. _____Costlines
108- --------- are found mainly at temperate latitudes. _____Kelp forests
109- -------- are limited to the trophic & subtrophic. _____Coral reefs
110- There are ---- types of coral reefs, frining reef, barrier reef, atolls. _____ 3
111- Kelp forests limited to areas with temperature range --------. _____10-20℃

112- Reef building corals limited to areas with temperature-------. _____18-29℃
113- Tidal fluctuation produce -------- due to gravitational effects of sun and moon. _____Steep gradience
114- Sult marshes dominated by -------- vegetation. _____Herbaceous
115- --------- grows in the tropic and subtropics. _____Mangrove forests
116- --------are extremely dynamic physically, chemically and biologically. _____Estuaries
117- ---------- are much like small seas. _____Lakes
118- The salinity of lakes ranges from very dilute waters to over -------. _____200%
119- Freshwater environment are particulary vulnerable to introduction of ------- ._____ Exotic species
120- The nature of fish assemblage is used to assess the ------- of fresh water. _____Biological integrity
121- For the individual organism macroclimate may be ------- significant than microclimate. _____Less
122- Most species perform best in a fairly -------- of temperature. _____Narrow range
123- The influence of temperature on the performance of organisms begins at the -------. _____Molecular level
124- Plants and ectothermic animals use morphology and behavior. _____Both
125- Birds and mammals rely heavily on --------- to regulate body temperature. _____Metabolic energy
126- Most endothermic aquatic species are ----------. _____Air breathers
127- Flying insects & marine fish improve performance by heating parts of their ----------. _____Anatomy
128- The energetic requirements of thermoregulation may influence the -------- distribution of species. _____Geographic
129- -------- is a state of low metabolic rate and low body temperature when food is scarce and night temperature cold. _____Torpor
130- Hibernation occur in winter, if it occur in summer then called -------. _____Aestivation
131- The most familiar relative measure of the water content of air is relative -------. Humidity
132- In --------- environment water moves from hypo osmotic to hyper osmotic solution. _____Aquatic
133- Larger osmotic difference b/w organism and environment generate --------. _____High osmotic pressure
134- The water potential of pure water is set at ----------. _____Zero
135- Water potential, vapor pressure and osmotic pressure are all measured in ----------. _____Pascals ( $N/M^2$)
136- Aquatic organisms gain or lose water through ----------. _____Osmosis
137- There are ---- source of energy. Light, organic & inorganic molecules. _____3
138- Photosynthetic plants and algae use light of wavelength b/w ------. _____
a. 400-700nm
139- PAR stands for ----------._____ Photosynthetic active radiations
140- There are----major alternative photosynthetic pathways,C3,C4 & CAM. _____ 3
141- Heterotrophs use organic molecules as a source of carbon &----._____Energy
142- ------- feed on dead plants material with lower nitrogen. _____Detritivores
143- Chemosynthetic bacteria use ------ molecules._____ Inorganic
144- Bacteria are most diverse in the --------._____ Biosphere

145- **The relationship b/w food density and animal feeding rate is called -----.**_____ **Functional response**

146- **Energy limitation is a assumption of ------- theory.** _____**Optimal fraging**

147- ……alters the outcome of competition by removing nutrient limitation on stress-tolerant plants _____ **Fertilization**

148- There Are **Four** Possible Outcomes of Interspecific Competition

149- PAR, accounts for about **45%** of the total energy

150- osmotic pressure, and vapor pressure can be measured in **Pascal's or (Newton's/m2)**

151- **Sociality** is generally accompanied by cooperative feeding, defense of the social group, and restricted reproductive opportunities

152- In the logistic model, the rate of population growth **decreases** as population density increases.

153- A **landscape** is a heterogeneous area composed of several ecosystems

154- **Exploitation** is an interaction between species that enhance the fitness of one individual at the expense of another

155- The pathogenic microbes such as moth larva reduced the coverage of prickly pear cactus in **Australia**

156- **Diarrhea and dysentery** caused by an amoeba

157- A **mutation** is any change in the proper nucleic acid sequence of a specific gene in a cell's genome

158- Ecology is derived from **Greek** word.

159- **Odum** --- "The structure and function of Nature"

160- A **polyphyletic** grouping consists of various species that lack a common ancestor

161- Freshwater contain less than **1%** water

162- The experiment failed because the original cause **rats** had not been eliminated.

163- The notochord is a flexible, rod-like structure derived from **mesoderm**.

164- Zooplankton feed on----------? Options a) bacteria b) fungi c) algae d) all

165- Chinese farmers have used **ants** as keystone predators to control pests in citrus orchard

## BEST OF LUCK